

Blog on statistical mechanics

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1 Probabilistic Models in Science and the Role of $\mathbf{E}$

The motivation for this blog post is to understand and derive statistical-mechanics-type models in a way that is accessible to researchers in psychological network psychometrics.

The canonical form we want to understand is the following:

$$P(\mathbf{x}) = \frac{\exp\{x(\mathbf{x})\}}{Z}, \quad Z = \sum_{\mathbf{x}} \exp\{x(\mathbf{x})\},$$

As seen in the Ising, Potts, and Blume–Capel families.

These models have been wildly successful in physics at connecting the *macro-level* we experience to the *micro-level* of atoms. For instance, we experience the solid, fluid, and gaseous phases of water, and how there are

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sudden changes in the phases when water reaches certain temperatures. The great success of statistical mechanics is to explain these phase transitions through the influence of temperature on the atoms.

This has not been done through massive simulation of gazillions of atoms. Rather, it has been achieved by using statistical laws to obtain relatively simple models. The models have also revealed *universality* classes—materials that share macroscopic behavior, despite chemical differences, due to their similar underlying structure described by statistical-mechanics models.

In the rest of this blog, we derive their form from first principles based on (?) and Leonard Susskind’s lectures on statistical mechanics¹.

1.1 Building Probabilistic Models and Why Always e^x

The *core assumption* of statistical mechanics is that **the probability of a state is proportional to the number of ways it can occur**. If many roads lead to Rome while only one way leads to Grenen, Denmark, then the random hitchhiker is much more likely to pass by Rome.

First, we introduce our mathematical objects. We are interested in systems that are made up of many identical elements that interact. In physics, these are typically atoms, while in psychology they can be belief elements (feelings, thoughts, behaviors) or depression symptoms.

The system is made of N elements, where each element can be in a number of states. Atoms can spin up or down, while symptoms can be absent or present (more than 2 states are also possible).

We now introduce the *occupation number*, n_i , which is the number of elements in state i . For example, in the depression case there are two states ($K = 2$): *present* and *absent*. Thus $n_{\text{present}} + n_{\text{absent}} = N = 9$.

A macro-state is defined by a full set of occupation numbers $(n_1, n_2, \dots, n_K)$, i.e., how many elements are in each of the K states. We can compute in how many ways a macro-state can occur as follows:

$$c_n = \frac{N!}{n_1! n_2! \cdots n_K!}. \quad (1)$$

In the depression example with $K = 2$, the macro-state can be described by a single number $m = n_{\text{present}}$ (since $n_{\text{absent}} = N - m$). Equation 1 then reduces to the binomial coefficient

$$c_m = \frac{N!}{m! (N - m)!}. \quad (2)$$

¹<https://www.youtube.com/watch?v=MCRHXHskWnE>

For example:

$$c_9 = \frac{9!}{9!0!} = 1, \quad c_7 = \frac{9!}{7!2!} = 36.$$

In Table 1, we list how many ways each depression sum score can occur. Under our simplifying assumption, the probability of macro-state n is just its share of ways it can occur:

$$P(n) = \frac{c_n}{\sum_{m=0}^N c_m}. \quad (3)$$

We report these probabilities in the rightmost column of Table 1. Equation 3 captures the core idea of statistical mechanics: a macro-state’s probability is its fraction of all possible realizations—the number of ways it can occur relative to every other macro-state. Additionally, we see that some macrostates are more likely than others: in general, Equation 1 (and thus c_n) is maximised when $n_1 = n_2 = \dots = n_K$, i.e., when elements are uniformly distributed across states. For the depression example with $N = 9$, we see that $P(4) = P(5) \approx 0.246$.

n (present)	c_n	$P(n)$
0	1	0.001953125
1	9	0.01757812
2	36	0.0703125
3	84	0.1640625
4	126	0.2460938
5	126	0.2460938
6	84	0.1640625
7	36	0.0703125
8	9	0.01757812
9	1	0.001953125

Table 1: Number of ways in which depression sum scores (macrostates n) can occur and their probability.

1.2 From Combinatorics to Entropy

The concept of entropy plays a central role in statistical mechanics. Here I want to demystify it by showing that *maximising the number of ways a macro-state can occur is the same as maximising its entropy*. Along the way, we’ll see why e^x keeps appearing.

Equation 1 counts how many ways macro-states can occur. The problem is that it contains factorials, which grow extremely fast and are hard to

handle in closed form. We therefore simplify using Stirling's approximation:

$$\log(N!) \approx N \log N - N \quad \Leftrightarrow \quad N! \approx N^N e^{-N}.$$

Crucially, the e^x pops out naturally—this is why it shows up *everywhere* in statistical mechanics. Using the second form of Stirling's approximation in Equation 1, we get

$$c_n = \frac{N!}{n_1! n_2! \cdots n_K!} \approx \frac{N^N e^{-N}}{\prod_{i=1}^K (n_i^{n_i} e^{-n_i})} = \frac{N^N e^{-N}}{\left(\prod_{i=1}^K n_i^{n_i}\right) e^{-\sum_{i=1}^K n_i}} = \frac{N^N}{\prod_{i=1}^K n_i^{n_i}}, \quad (4)$$

where we used $\sum_{i=1}^K n_i = N$ to cancel the exponentials.

We then take logarithms because we will optimise this quantity later:

$$\log c_n \approx N \log N - \sum_{i=1}^K n_i \log n_i.$$

Now write $n_i = NP_i$, where P_i is the fraction of elements in state i (so $\sum_{i=1}^K P_i = 1$). Substituting,

$$\begin{aligned} \log c_n &\approx N \log N - \sum_{i=1}^K (NP_i) \log(NP_i) \\ &= N \log N - \sum_{i=1}^K NP_i (\log N + \log P_i) \\ &= N \log N - N \log N - N \sum_{i=1}^K P_i \log P_i \\ &= -N \sum_{i=1}^K P_i \log P_i. \end{aligned}$$

Thus, maximising the number of ways a macro-state can occur (equivalently, maximising $\log c_n$) is the same as *maximising the Shannon entropy*

$$S(P) = - \sum_{i=1}^K P_i \log P_i.$$

We arrived at this by applying Stirling's approximation and a log transformation. Entropy is therefore a mathematically convenient quantity whose maximisation selects the macrostates that can occur in the most ways.

In Table 2, we extend the previous table by calculating c_n using Stirling's approximation and the corresponding probabilities $P(n) = c_n / \sum_{m=0}^N c_m$. In the last column, we report the differences between exact and Stirling-based probabilities. See Appendix 3 for code.

n_{present}	c_n	$P(n)$	c_n^{Stirling}	$P^{\text{Stirling}}(n)$	$P(n) - P^{\text{Stirling}}(n)$
0	1	0.002	1.000	0.002	0.000
1	9	0.018	9.771	0.018	-0.001
2	36	0.070	37.619	0.071	-0.001
3	84	0.164	86.757	0.164	0.000
4	126	0.246	129.601	0.245	0.001
5	126	0.246	129.601	0.245	0.001
6	84	0.164	86.757	0.164	0.000
7	36	0.070	37.619	0.071	-0.001
8	9	0.018	9.771	0.018	-0.001
9	1	0.002	1.000	0.002	0.000

Table 2: Comparison of exact probabilities $P(n)$ and Stirling-approximate probabilities $P^{\text{Stirling}}(n)$, along with differences in counts and probabilities.

2 From Entropy to the e^x/Z Form

Many statistical-mechanics models end up looking like

$$p(i) = \frac{e^{x_i}}{Z} \quad \text{with} \quad Z = \sum_k e^{x_k}.$$

Where does this come from? Remember: by maximising entropy we select the macrostates that can occur in the most possible ways. We now apply that “most ways” logic to find an *entire distribution* over configurations. That is, we specify some constraints on our distribution like normalisation, $\sum_i P_i = 1$, correlations between variables, or an average value across variables. We then identify the distribution that can occur in the most ways and that satisfies the constraint.

Case 1: Only normalisation

To illustrate this logic, we start with a simple example. We want the max-entropy distribution where probabilities sum to 1. We can solve this problem using Lagrangian multipliers (a method for optimisation with constraints). If we include only the constraint $\sum_i P_i = 1$, the Lagrangian becomes:

$$\mathcal{L}(P, \alpha) = - \sum_i P_i \log P_i + \alpha \left(\sum_i P_i - 1 \right).$$

Differentiate with respect to P_i and set to zero:

$$\frac{\partial \mathcal{L}}{\partial P_i} = -(\log P_i + 1) + \alpha = 0 \implies \log P_i = \alpha - 1.$$

Exponentiate to solve for P_i :

$$P_i = e^{\alpha-1}.$$

This shows P_i does not depend on i , so all P_i are equal. Enforce normalization:

$$1 = \sum_{i=1}^K P_i = \sum_{i=1}^K e^{\alpha-1} = K e^{\alpha-1} \implies e^{\alpha-1} = \frac{1}{K}.$$

Substitute back to get the solution:

$$P_i = \frac{1}{K}, \quad i = 1, \dots, K.$$

This makes sense: if we know nothing except that probabilities must sum to one, the maximum-entropy distribution is uniform. In our depression example with $K = 2$ states (present/absent), this reads $P(\text{present}) = P(\text{absent}) = \frac{1}{2}$.

Case 2: Adding interactions, the pairwise Ising model

To move beyond the trivial baseline, let's assume we have measured the correlation between nodes. The basic "alignment" feature is $s_i s_j$, which is $+1$ when i and j are in the same state (both present or both absent) and -1 when they differ.

We then constrain the expected pairwise alignments based on our empirically observed correlation, C_{ij} :

$$\sum_{\mathbf{s}} P(\mathbf{s}) s_i s_j = C_{ij} \quad \text{for each pair } (i, j),$$

and introduce one Lagrange multiplier per potential edge, J_{ij} . Intuitively, J_{ij} becomes the *edge weight*.

We now form the Lagrangian that asks us to find the distribution with max entropy, probabilities summing to 1, and constrained interactions between variables:

$$\mathcal{L}(P, \alpha, J) = - \sum_{\mathbf{s}} P(\mathbf{s}) \log P(\mathbf{s}) + \alpha \left(\sum_{\mathbf{s}} P(\mathbf{s}) - 1 \right) + \sum_{i < j} J_{ij} \left(\sum_{\mathbf{s}} P(\mathbf{s}) s_i s_j - C_{ij} \right),$$

where $\mathbf{s} = (s_1, \dots, s_N)$ is a full configuration (microstate).

The full derivation mirrors the max-entropy calculus we used before, but now at the level of configurations with one multiplier per interaction. See Appendix 3 for the full derivation. The result is:

$$P(\mathbf{s}) \propto \exp \left\{ \sum_{i < j} J_{ij} s_i s_j \right\},$$

and, after normalising over all configurations, the *pairwise Ising network*:

$$P(\mathbf{s}) = \frac{1}{Z} \exp\left\{\sum_{i<j} J_{ij} s_i s_j\right\}, \quad Z = \sum_{\mathbf{s}} \exp\left\{\sum_{i<j} J_{ij} s_i s_j\right\}.$$

To summarize: Thus we arrive at the Ising model by showing that it assigns probabilities to micro stats such that the observed co-occurrences between nodes can occur in the most number of ways.

3 Code

```

1 # Load necessary libraries
2 library(dplyr)
3
4 # Function to calculate factorial
5 factorial <- function(x) ifelse(x == 0, 1, prod(1:x))
6
7 # Number of elements (N) and the possible states
8 N <- 9
9 states <- 0:N
10
11 # Calculate original c_n using factorial
12 c_n <- sapply(states, function(n) factorial(N) / (factorial(n)
13   * factorial(N - n)))
14
15 # Compute probabilities of macro-states
16 P_n <- c_n / sum(c_n)
17
18 # Stirling's approximation for factorial
19 stirring_approximation <- function(x) {
20   if (x == 0) return(1)
21   sqrt(2 * pi * x) * (x / exp(1))^x
22 }
23
24 # Calculate c_n using Stirling's approximation
25 c_n_stirling <- sapply(states, function(n) {
26   stirring_approximation(N) /
27   (stirling_approximation(n) * stirring_approximation(N - n))
28 })
29
30 # Compute probabilities using Stirling's approximation
31 P_n_stirling <- c_n_stirling / sum(c_n_stirling)
32
33 # Calculate entropy using original probabilities
34 entropy <- -sum(P_n * log(P_n))
35
36 # Calculate entropy using Stirling's approximation

```

```

36 entropy_stirling <- -sum(P_n_stirling * log(P_n_stirling))
37
38 # Compare results in a data frame
39 comparison <- data.frame(
40   n_present = states,
41   c_n = c_n,
42   P_n = P_n,
43   c_n_stirling = c_n_stirling,
44   P_n_stirling = P_n_stirling,
45   difference_c_n = c_n - c_n_stirling,
46   difference_P_n = P_n - P_n_stirling
47 )
48
49 # Add entropy to the results
50 entropy_comparison <- data.frame(
51   Metric = c("Entropy", "Entropy (Stirling)"),
52   Value = c(entropy, entropy_stirling)
53 )
54
55 # Output the comparison table
56 print("Comparison of probabilities and counts:")
57 print(round(comparison,3))

```

Listing 1: R script for comparing exact and Stirling’s approximation calculations.

Appendix: Max–Entropy with Interaction Constraints (Derivation)

Setup. Variables $s_i \in \{-1, +1\}$, configuration $\mathbf{s} = (s_1, \dots, s_N)$, unknown distribution $P(\mathbf{s})$. Constraints: (i) normalisation $\sum_{\mathbf{s}} P(\mathbf{s}) = 1$; (ii) expected pair alignments $\sum_{\mathbf{s}} P(\mathbf{s}) s_i s_j = C_{ij}$ for each (i, j) .

Lagrangian.

$$\mathcal{L}(P, \alpha, J) = - \sum_{\mathbf{s}} P(\mathbf{s}) \log P(\mathbf{s}) + \alpha \left(\sum_{\mathbf{s}} P(\mathbf{s}) - 1 \right) + \sum_{i < j} J_{ij} \left(\sum_{\mathbf{s}} P(\mathbf{s}) s_i s_j - C_{ij} \right).$$

Stationary conditions. Differentiate w.r.t. $P(\mathbf{s})$ and set to zero:

$$\frac{\partial \mathcal{L}}{\partial P(\mathbf{s})} = -(\log P(\mathbf{s}) + 1) + \alpha + \sum_{i < j} J_{ij} s_i s_j = 0.$$

Solve for $P(\mathbf{s})$. Rearrange:

$$\log P(\mathbf{s}) = \alpha - 1 + \sum_{i < j} J_{ij} s_i s_j \implies P(\mathbf{s}) = e^{\alpha-1} \exp\left\{ \sum_{i < j} J_{ij} s_i s_j \right\}.$$

Normalisation (partition function). We must choose α so that probabilities sum to one:

$$\sum_{\mathbf{s}} P(\mathbf{s}) = 1.$$

With $P(\mathbf{s}) = e^{\alpha-1} \exp\{\sum_{i < j} J_{ij} s_i s_j\}$ this gives

$$e^{\alpha-1} \sum_{\mathbf{s}} \exp\left\{ \sum_{i < j} J_{ij} s_i s_j \right\} = 1.$$

Define the *partition function*

$$Z := \sum_{\mathbf{s}} \exp\left\{ \sum_{i < j} J_{ij} s_i s_j \right\}.$$

Then $e^{\alpha-1} = 1/Z$ and the normalised model is

$$P(\mathbf{s}) = \frac{1}{Z} \exp\left\{ \sum_{i < j} J_{ij} s_i s_j \right\}.$$